



Forest Disturbance Monitoring



Forest mapping



🌲 Forest mapping - Not so easy

Forest **definition**

Tree cover %, tree height, species, land use

Forests are **dynamic systems**

Natural growth, (un)planned disturbances

Mapping forests is **non-trivial**

Spatiotemporal scale, biomes, sensors, semantics





Forest disturbance monitoring



Pests



Fire



Windthrow



Logging

In-situ observation accurate but not scalable - need for **EO-based monitoring**

Various event speeds, requiring **daily to monthly monitoring**

Diverse **agents** affecting **tree cover** and **land use** and potentially **correlating**

Require temporal understanding for **change detection** and **agent attribution**

Regional patterns requiring tailored models for local forest practitioners



Approach - Existing products

	Existing works	Desiderata
Latency	Yearly, few near real-time	Near real-time
Output	Detection XOR attribution	Detection AND attribution
Modality	Cloud-free optical XOR SAR	Optical AND SAR
Model	Heuristics, no spatial context, non-DL	Spatio-temporal DL
Tuning	Region-specific	Tunable to small regional datasets
Reference	Few 1x1 Landsat pixels, no agents	DL-friendly benchmark
Inference	GEE-aided, no DL at scale	Open DL-based system

Benchmarking dataset





Reference data

 **3,777** sample locations in **Europe**

 **2015-2024**

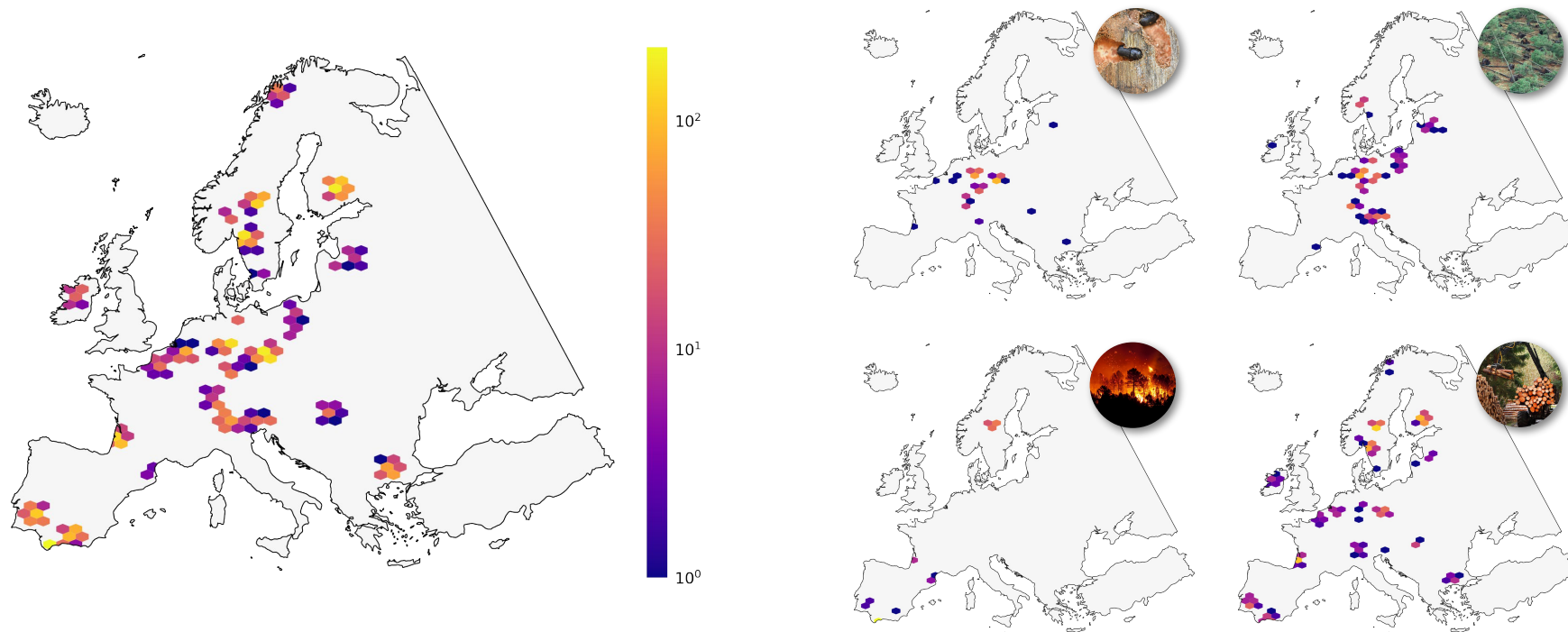
 **+2k** event / **+5k** segment annotations -    

 **252x252** pixel patches

 **5M Sentinel-1** GRD patches

 **800k Sentinel-2** L2A cloud-free patches

 **2TB** storage



Geographic distribution



Reference data



3,777 sample locations in **Europe**



2015-2024



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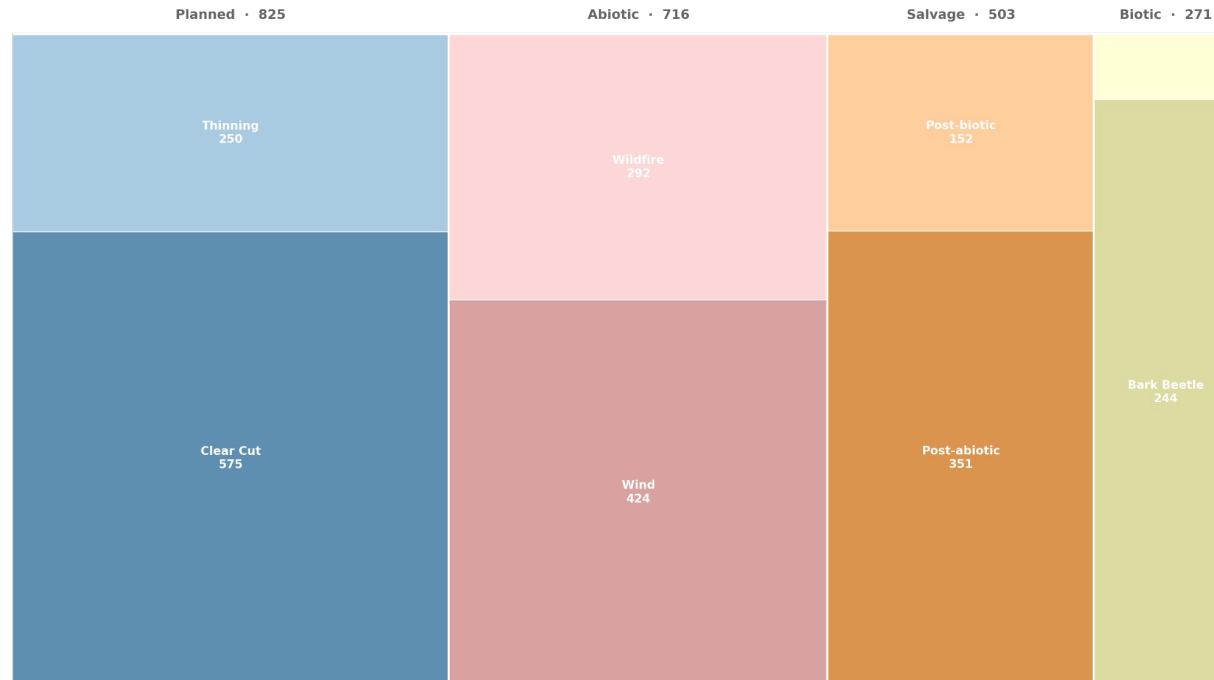
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Agent distribution



Reference data

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 **2015-2024**

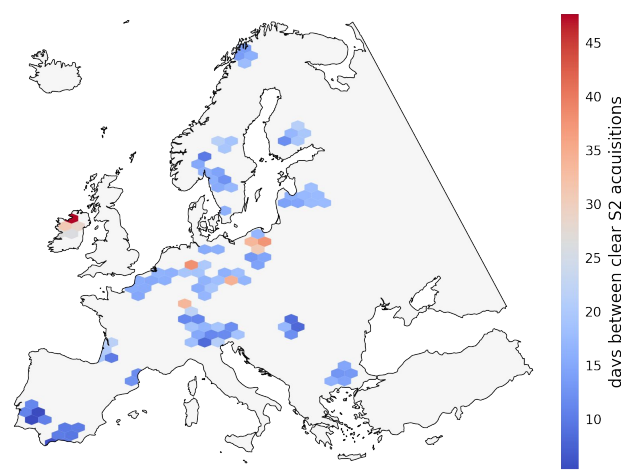
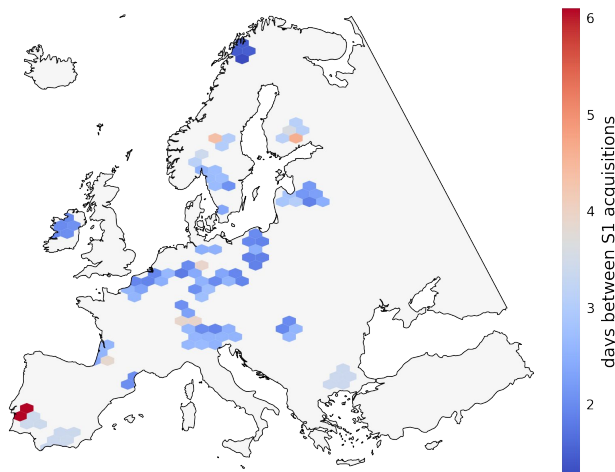
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Sentinel-1/2 revisit

Sample time series

 **3,777** sample locations in **Europe**

 **2015-2024**

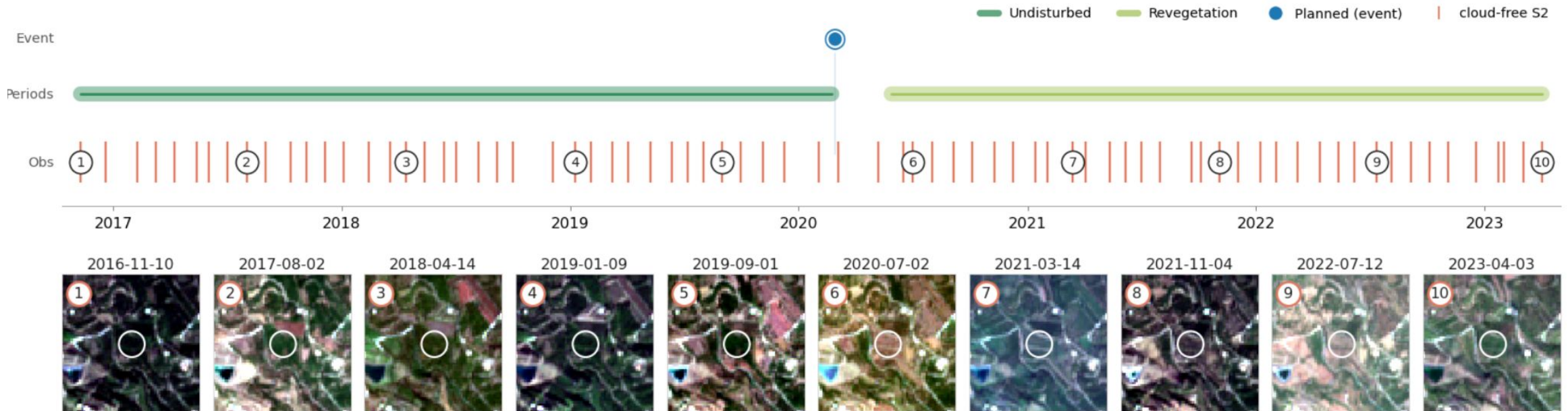
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 Annotations **only on center pixel** (126, 126)



Sample time series



3,777 sample locations in **Europe**



2015-2024



+2k event / **+5k** segment annotations -    



252x252 pixel patches



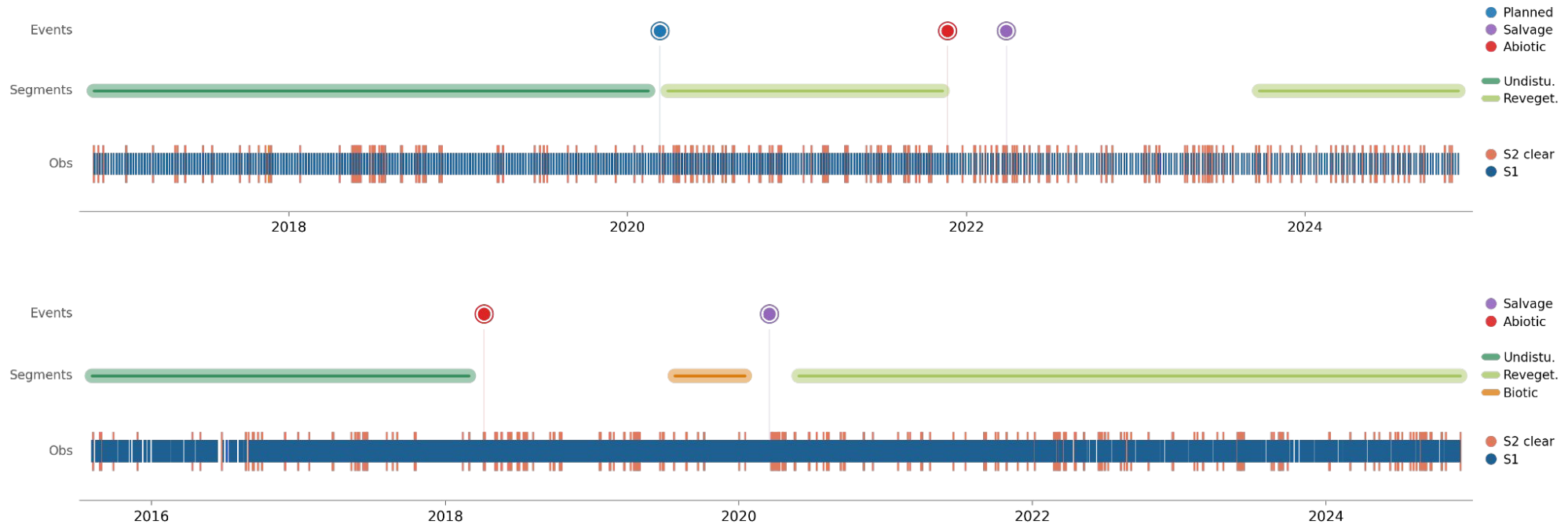
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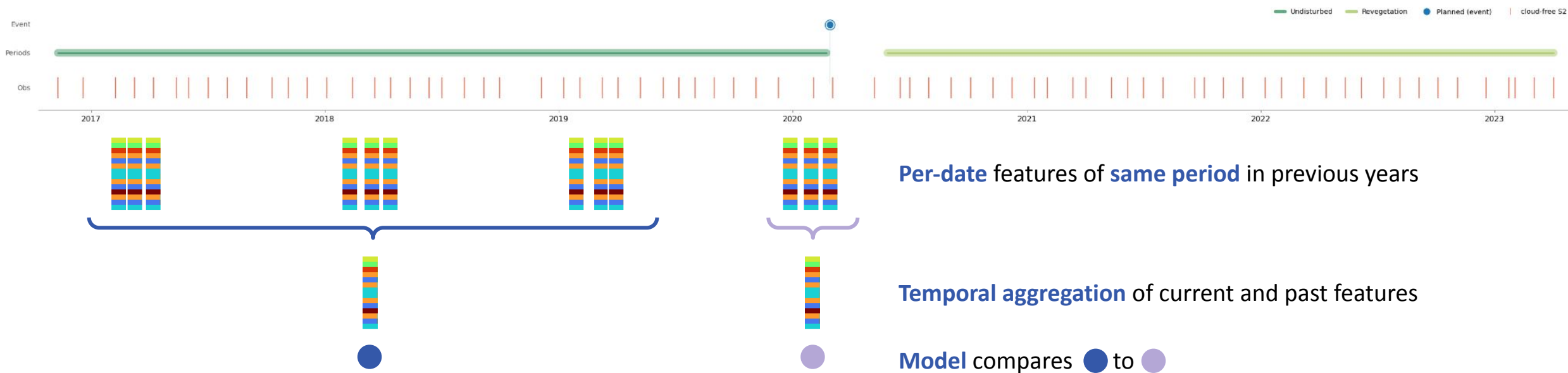
Complex vegetation histories



Benchmarking tasks



Modeling: inputs



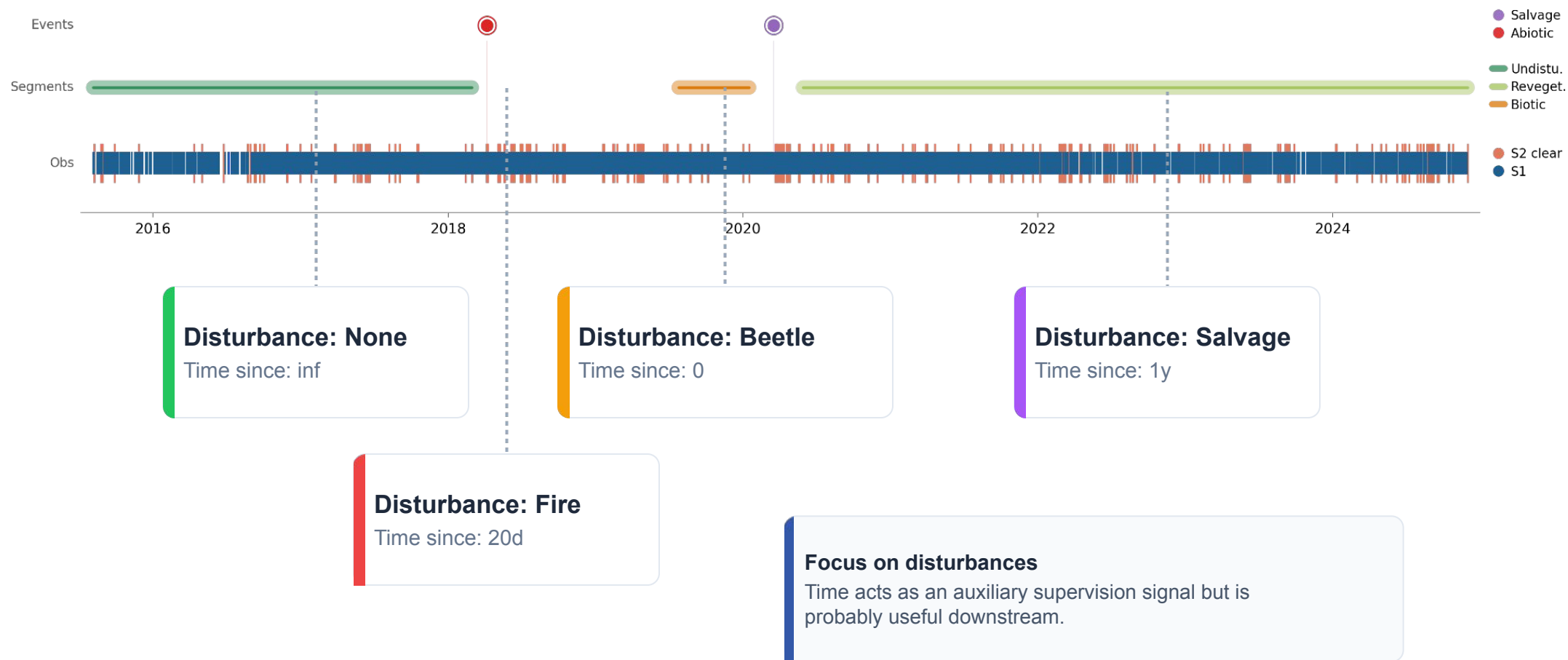
Near real-time detection task

Near real-time attribution task

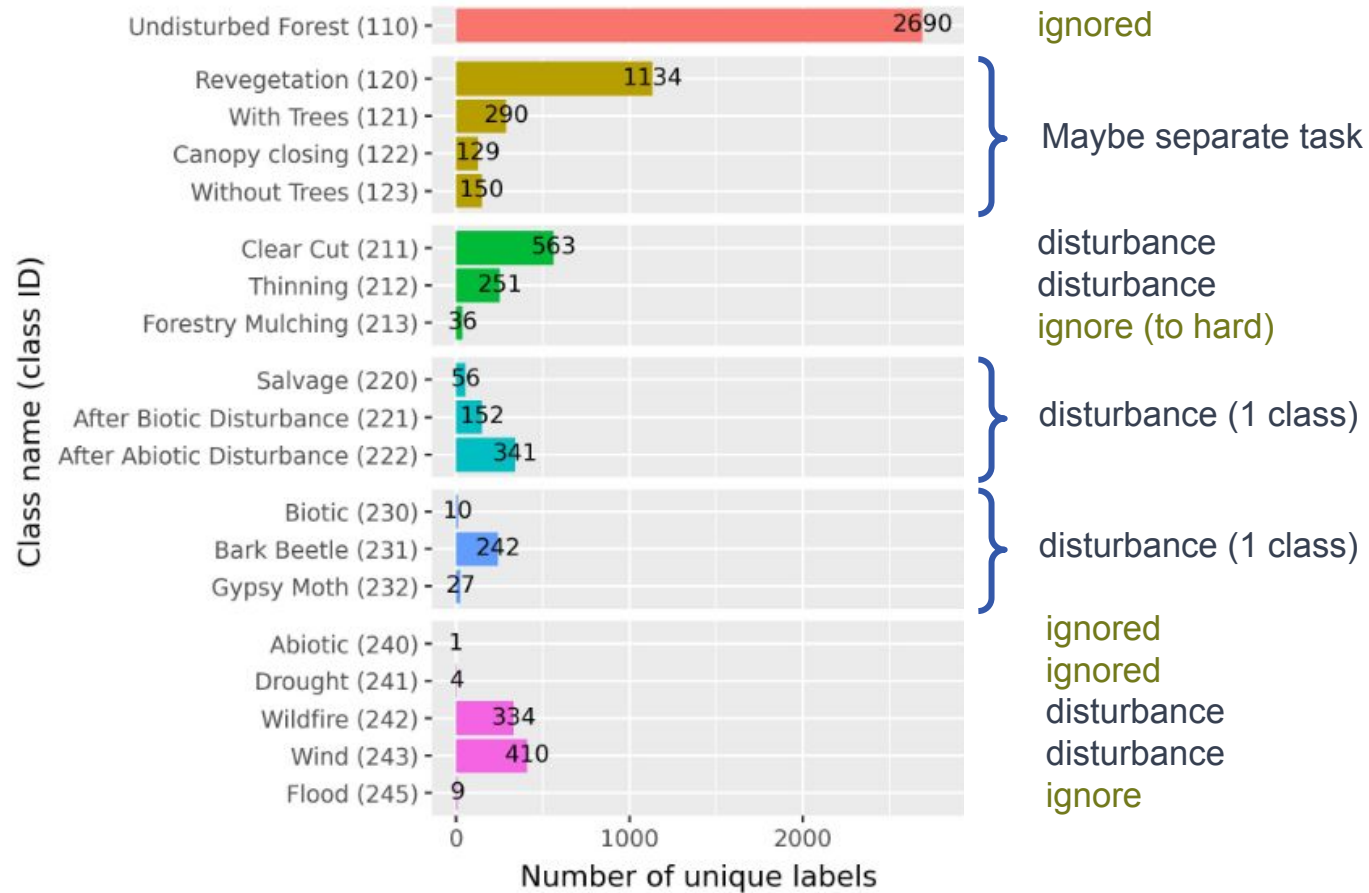
Disturbance at ● ? 🚒

🪓, ✂️, 🔥, or 🧑‍🔧 between ● and ● ?

Modeling: labels



Modeling: labels



Modeling: loss

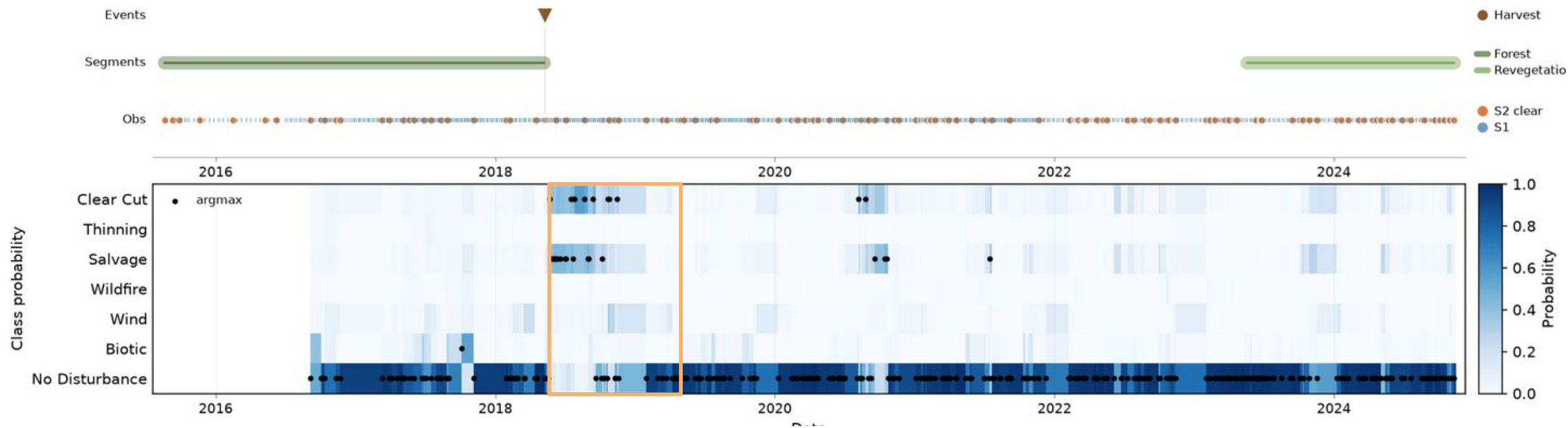
1. Disturbance classification

- Uses **cross-entropy** on disturbance ID.
- **Soft Forget logic**: if duration exceeds forget period, prediction tends towards no disturbance class ID.
- No disturbance class ID samples still contribute to loss.

2. Elapsed time regression

- Uses **L2** on time since last event. (maybe on inv?)
- **Step ignored** for invalid, ignored, or unfinished targets (like inf without past disturbance).

Evaluation: metrics



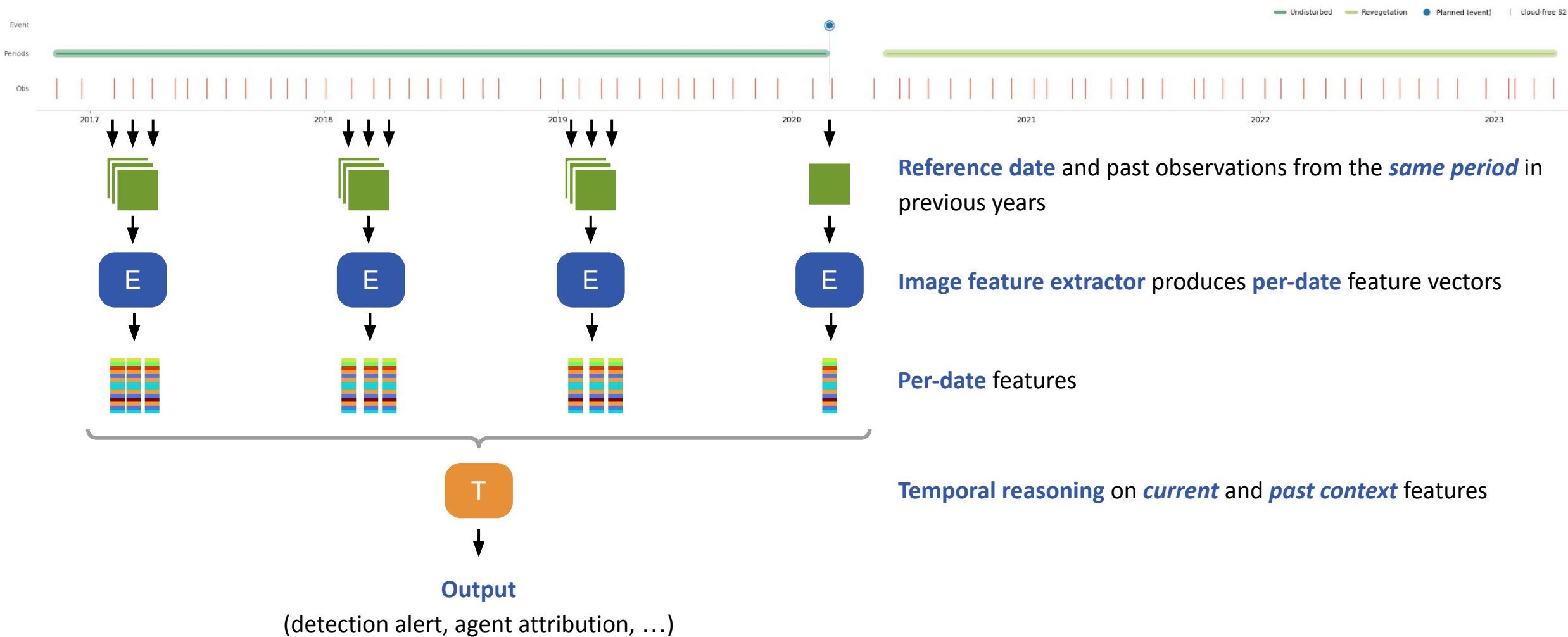
Event based

Correct prediction if **at least one alert** in the **horizon**, otherwise false
negatif
→ Event-based recall

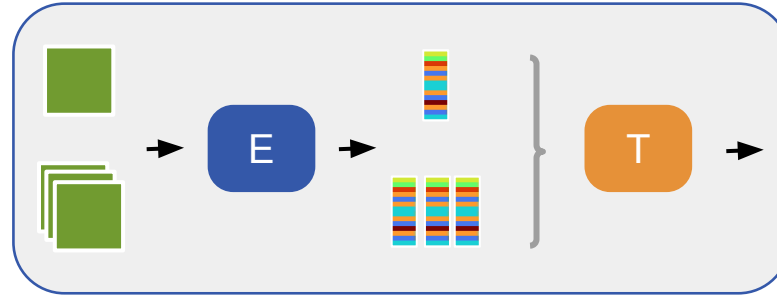
Alert based

Alert is correct if in the **horizon**, otherwise false positive
→ Alert-based precision

Modeling: general pipeline



Modeling: a simple “MLP” baseline



E

`MLP([sentinel-2 bands, NDVI, handcrafted indices, ...])`

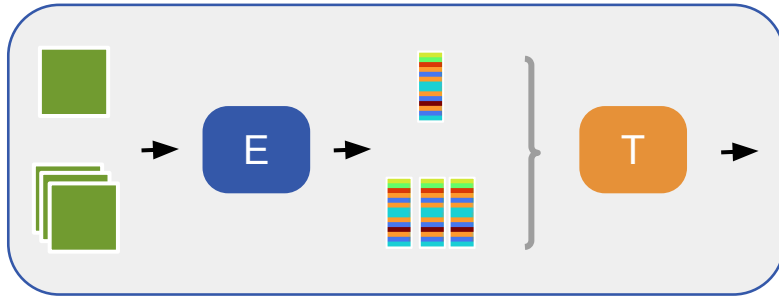
T

`MLP([mean_past_feats, std_past_feats, current_feats])`



A basic approach with lots of possible improvements

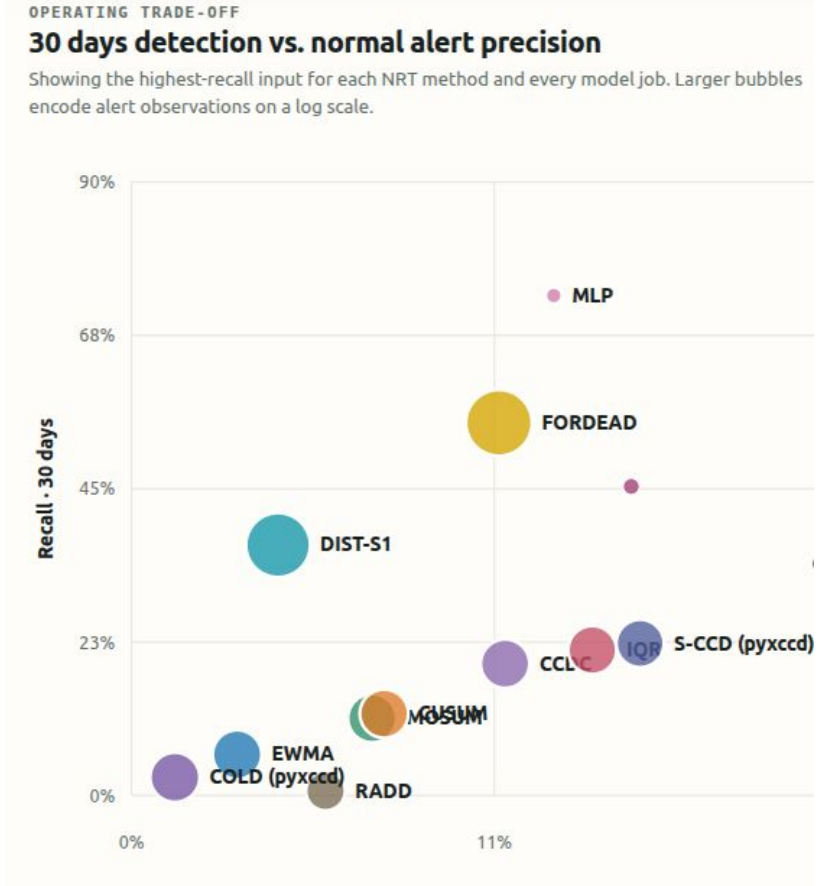
🎯 Modeling: a simple “MLP” baseline



E `MLP([sentinel-2 bands, NDVI, handcrafted indices, ...])`

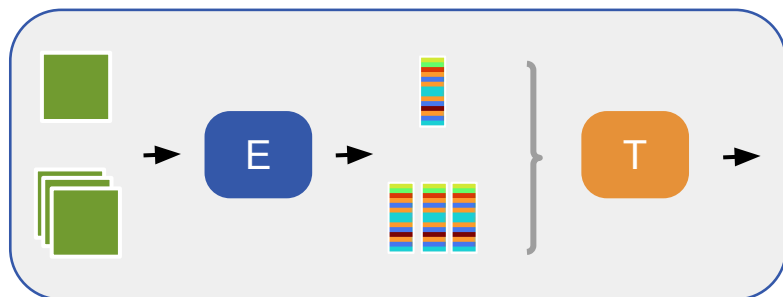
T `MLP([mean_past_feats, std_past_feats, current_feats])`

💡 A basic approach with lots of possible improvements



MLP baseline outperforms established methods

🎯 Modeling: ongoing research ideas

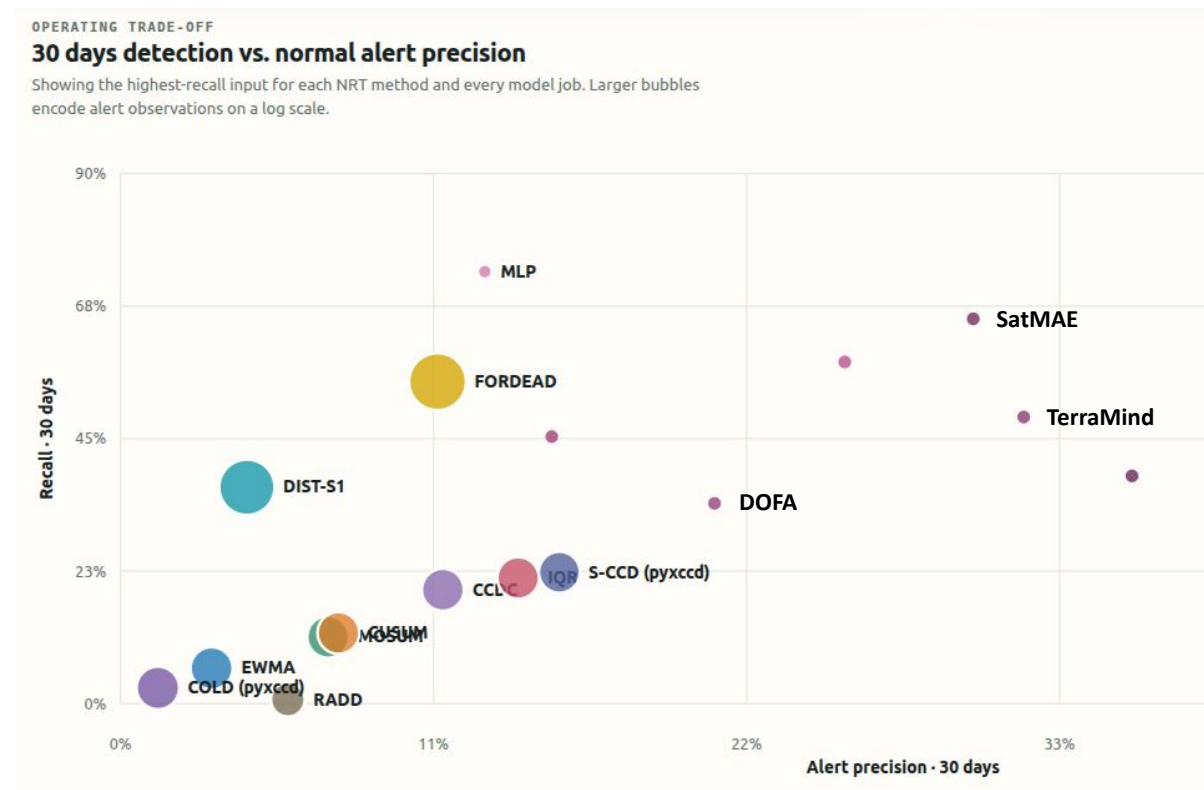


E

$\text{MLP}(\mathbf{F}([\text{sentinel-2 bands}]))$

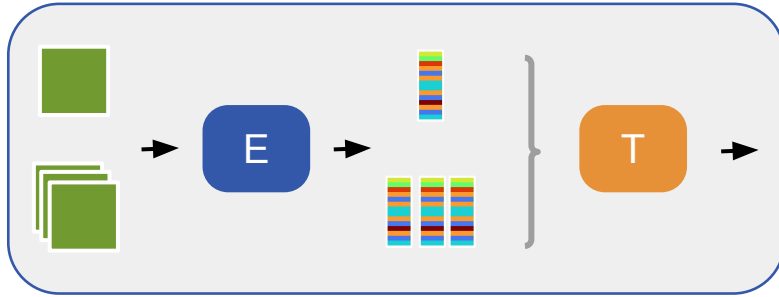
Leverage model $\mathbf{F}$ pretrained on other related tasks:

- foundation models: SatMAE, DOFA, TerraMind, ...
- forest models: canopy height / biomass regression, ...
- embedding products : AlphaEarth, ... (*often yearly!*)



Promising results - ongoing research

Modeling: ongoing research ideas



 `T([past_feats, current_feats])`

Leverage more advanced time series modeling **T**:

- LSTM
- GRU
- Transformer
- Mamba
- POM



For operational deployment, recurrent models would be ideal...

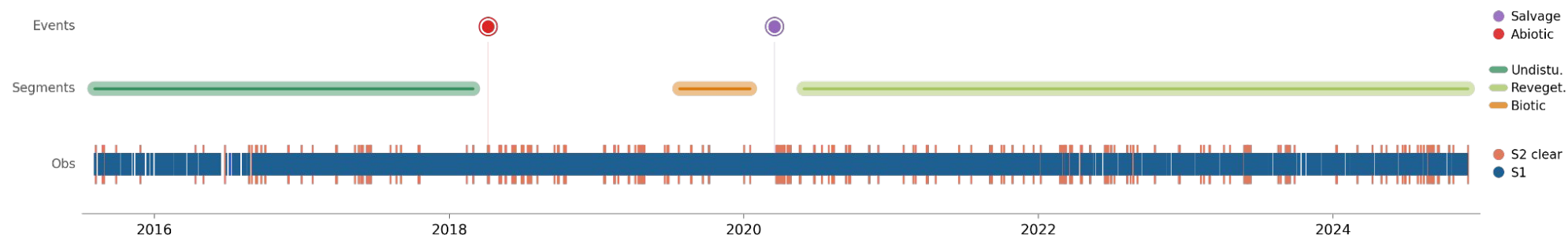


Next steps & discussion





What's next



Learning paradigm

- Only **3.7k** time series
- Only **x1 center pixel** annotations
- **Event** vs **segment** annotations
- **Rare** classes



Loss and metrics

- **Early** detection and attribution
- **Hierarchical** agent attribution
- **Temporal gaps** in annotations



Modelling

- **NRT data loading** paradigm
- **Temporal reasoning head**
- **Multimodal** learning
- Temporal uncertainty **refinement**

Benchmarking

- Prior forest disturbance models
- Deep learning baseline from scratch
- Geospatial foundation models
- Forest-related backbones

Inference

- Alert system deployment
- Inference at scale
- Rigorous product assessment